

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I Sweta R (192511408) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Sweta R

Annexure A

Analytical Problem Solving

1. RSA Encryption Analysis

Given $p=11$, $q=13$

- a. Compute n and $\phi(n)$
- b. Choose $e=7$ and find the private key d
- c. Encrypt the message $M=9$
- d. Decrypt the ciphertext and verify the result
- e. Analyze how changing e affects security

Solution:

Given:

- $p = 11$
- $q = 13$
- $e = 7$
- $M = 9$

a. Compute n and $\phi(n)$

$$\begin{aligned}n &= p \times q \\&= 11 \times 13 \\&= 143\end{aligned}$$

$$\begin{aligned}\phi(n) &= (p - 1)(q - 1) \\&= (11 - 1)(13 - 1) \\&= 10 \times 12 \\&= 120\end{aligned}$$

Answer:

- $n = 143$
- $\phi(n) = 120$

b. Choose $e = 7$ and find the private key d

Find d such that:

$$7 \times d \equiv 1 \pmod{120}$$

Using the Extended Euclidean Algorithm,

$$d = 103$$

Private Key = (103, 143)

c. Encrypt the message M = 9

Encryption formula:

$$C = M^e \bmod n$$

$$C = 9^7 \bmod 143$$

$$9^7 = 4,782,969$$

$$4,782,969 \bmod 143 = 48$$

Ciphertext = 48

d. Decrypt the ciphertext and verify the result

Decryption formula:

$$M = C^d \bmod n$$

$$M = 48^{103} \bmod 143$$

$$M = 9$$

Verified: The decrypted message is the same as the original message.

e. Analyze how changing e affects security

- The value of e must be relatively prime to $\phi(n)$.
- A smaller value of e provides faster encryption but may reduce security if used improperly.
- A larger value of e increases encryption time.
- A commonly used value is 65537 because it provides a good balance between security and efficiency.
- Choosing an appropriate value of e improves the overall security of the RSA algorithm.

2. Block Cipher Mode Evaluation

Given plaintext blocks:

$P = [1010, 1010, 1111, 1010]$, Key = $K = 1100$

- Encrypt using ECB and CBC modes (Assume IV = 0000)
- Compare the ciphertext outputs
- Analyze which mode hides patterns better and explain why

solution:

Given:

- Plaintext Blocks:
 - $P1 = 1010$
 - $P2 = 1010$
 - $P3 = 1111$
 - $P4 = 1010$
- Key (K) = 1100
- IV = 0000

Assumption: Encryption is performed using XOR operation.

a. Encrypt using ECB and CBC Modes

ECB (Electronic Code Book) Mode

$P1 = 1010 \text{ XOR } 1100 = 0110$

$P2 = 1010 \text{ XOR } 1100 = 0110$

$P3 = 1111 \text{ XOR } 1100 = 0011$

$P4 = 1010 \text{ XOR } 1100 = 0110$

ECB Ciphertext:

0110 0110 0011 0110

CBC (Cipher Block Chaining) Mode

Block 1

$P1 \text{ XOR } IV$

$1010 \text{ XOR } 0000 = 1010$

Encrypt:

$1010 \text{ XOR } 1100 = 0110$

$C1 = 0110$

Block 2

$P2 \text{ XOR } C1$

$1010 \text{ XOR } 0110 = 1100$

Encrypt:

$1100 \text{ XOR } 1100 = 0000$

$C2 = 0000$

Block 3

$P3 \text{ XOR } C2$

$1111 \text{ XOR } 0000 = 1111$

Encrypt:

$1111 \text{ XOR } 1100 = 0011$

$C3 = 0011$

Block 4

$P4 \text{ XOR } C3$

$1010 \text{ XOR } 0011 = 1001$

Encrypt:

$1001 \text{ XOR } 1100 = 0101$

$C4 = 0101$

CBC Ciphertext:

0110 0000 0011 0101

b. Compare the Ciphertext Outputs

Mode	Ciphertext
ECB	0110 0110 0011 0110
CBC	0110 0000 0011 0101

c. Analyze Which Mode Hides Patterns Better and Explain Why

- **ECB Mode** encrypts each block independently. Therefore, identical plaintext blocks produce identical ciphertext blocks, making patterns visible.
 - **CBC Mode** combines each plaintext block with the previous ciphertext block before encryption. As a result, identical plaintext blocks produce different ciphertext blocks.
 - **CBC Mode** hides patterns more effectively and provides better security than ECB mode.
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3. DES vs 3-DES vs Blowfish Performance Study

Given:

- a. DES key size = 56 bits, time = 2 ms/block
- b. 3-DES key size = 168 bits, time = 6 ms/block
- c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

- i. Calculate total encryption time for each algorithm
- ii. Analyze trade-offs between security and performance
- iii. Recommend the best algorithm for secure real-time communication

solution:

Given:

- DES key size = 56 bits, Time = 2 ms/block
- 3-DES key size = 168 bits, Time = 6 ms/block
- Blowfish key size = 128 bits, Time = 1 ms/block
- Number of data blocks = 1000

i. Calculate Total Encryption Time

DES

Time = 2 ms/block × 1000 blocks

= 2000 ms

= **2 seconds**

3-DES

Time = 6 ms/block × 1000 blocks

= 6000 ms

= **6 seconds**

Blowfish

Time = 1 ms/block × 1000 blocks

= 1000 ms

= **1 second**

ii. Analyze Trade-offs Between Security and Performance

Algorithm	Key Size	Total Time	Security	Performance
DES	56 bits	2 sec	low	good
3 DES	168 bits	6 sec	high	slow
Blowfish	128 bits	1 sec	high	excellent

Analysis:

- **DES** is the fastest after Blowfish but provides weak security due to its small 56-bit key.
- **3-DES** offers stronger security but takes the longest time because encryption is performed three times.
- **Blowfish** provides strong security with the shortest encryption time, making it highly efficient.

iii. Recommend the Best Algorithm for Secure Real-Time Communication

Recommended Algorithm: Blowfish

Reason:

- Provides strong security with a 128-bit key.
 - Requires only **1 second** to encrypt 1000 blocks.
 - Consumes less processing time than DES and 3-DES.
 - Suitable for secure real-time communication due to its high speed and strong encryption.
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4. Diffie-Hellman Key Exchange Analysis

Given:

- a. $p=23$, $g=5$
- b. User A private key = 6
- c. User B private key = 15
- d. Compute public keys for A and B
- e. Compute the shared secret key
- f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
- g. Suggest a secure method to prevent the attack

Solution:

Given:

- Prime number, **$p = 23$**
- Generator, **$g = 5$**
- User A private key = **6**
- User B private key = **15**

d. Compute Public Keys for A and B

User A Public Key

Public Key = $g^a \text{ mod } p$

$$= 5^6 \text{ mod } 23$$

$$= 15625 \text{ mod } 23$$

$$= 8$$

Public Key of A = 8

User B Public Key

$$\text{Public Key} = g^b \text{ mod } p$$

$$= 5^{15} \text{ mod } 23$$

$$= 19$$

Public Key of B = 19

e. Compute the Shared Secret Key

User A computes:

$$\text{Shared Key} = (\text{Public Key of B})^a \text{ mod } p$$

$$= 19^6 \text{ mod } 23$$

$$= 2$$

User B computes:

$$\text{Shared Key} = (\text{Public Key of A})^b \text{ mod } p$$

$$= 8^{15} \text{ mod } 23$$

$$= 2$$

Shared Secret Key = 2

f. Demonstrate How a Man-in-the-Middle (MITM) Attacker Could Alter the Exchange

- The attacker intercepts the public keys exchanged between User A and User B.
- The attacker replaces A's public key with its own public key before sending it to B.
- Similarly, the attacker replaces B's public key before sending it to A.
- As a result, User A and the attacker share one secret key, while User B and the attacker share another secret key.
- The attacker can read, modify, or forward messages without either user knowing.

g. Suggest a Secure Method to Prevent the Attack

- Use **Digital Signatures** to verify the identity of users.
 - Use **Digital Certificates** issued by a trusted Certificate Authority (CA).
 - Apply authenticated Diffie-Hellman protocols (such as Station-to-Station or TLS) to ensure secure key exchange.
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5. ECC vs RSA Comparison

Given:

- a. RSA key size = 1024 bits
- b. ECC key size = 160 bits (equivalent security)
- c. Device processing capacity = limited (IoT device)
- d. Analyze computational cost for both systems
- e. Estimate which consumes less power and memory
- f. Justify which cryptosystem is better for the given scenario with reasoning

solution:

Given:

- RSA key size = 1024 bits
- ECC key size = 160 bits (Equivalent security)
- Device processing capacity = Limited (IoT device)

d. Analyze Computational Cost for Both Systems

RSA:

- Uses a 1024-bit key.
- Requires more mathematical computations.
- Has a higher computational cost.
- Slower for encryption and decryption.

ECC:

- Uses a 160-bit key.
- Requires fewer mathematical computations.
- Has a lower computational cost.
- Faster than RSA while providing equivalent security.

Analysis:

ECC is computationally more efficient than RSA because it provides the same level of security with a much smaller key size.

e. Estimate Which Consumes Less Power and Memory

RSA:

- Consumes more power.
- Requires more memory.
- Not suitable for low-power devices.

ECC:

- Consumes less power.
- Requires less memory.
- Suitable for battery-powered and IoT devices.

Analysis:

ECC consumes less power and memory than RSA because of its smaller key size and lower computational requirements.

f. Justify Which Cryptosystem is Better for the Given Scenario

Recommended Cryptosystem: ECC (Elliptic Curve Cryptography)

Reason:

- Provides the same security as RSA with a smaller key size.
 - Requires less computational power.
 - Consumes less memory and battery power.
 - Offers faster encryption and decryption.
 - Ideal for IoT devices with limited processing capacity.
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RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

